

11.24

(a)

```
Call:
lm(formula = lprice ~ mon + tue + wed + thu + stormy + mixed,
    data = fultonfish)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.92281	-0.18819	0.00813	0.22404	0.68381

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	-0.35957	0.07915	-4.543	1.50e-05	***
mon	-0.10825	0.10339	-1.047	0.29753	
tue	-0.06609	0.10103	-0.654	0.51446	
wed	-0.04927	0.10377	-0.475	0.63591	
thu	0.03935	0.10071	0.391	0.69681	
stormy	0.44634	0.07921	5.635	1.51e-07	***
mixed	0.23688	0.07851	3.017	0.00321	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3413 on 104 degrees of freedom
 Multiple R-squared: 0.245, Adjusted R-squared: 0.2014
 F-statistic: 5.624 on 6 and 104 DF, p-value: 4.349e-05

∴ The p-value of mixed variable is 0.00321, It is statistically significant at the 5% level.

STORMY 與 MIXED 的聯合顯著性檢定:

```
> f_test_a <- linearHypothesis(model_a, c("stormy = 0", "mixed = 0"))
> print(f_test_a)
```

Linear hypothesis test:

stormy = 0

mixed = 0

Model 1: restricted model

Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	106	15.876				
2	104	12.115	2	3.7604	16.14	7.857e-07 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

H_0 : both stormy and mixed are zero vs H_1 : not H_0

$$F\text{-value} = 16.14 > 10$$

∴ p-value = $7.857 \times 10^{-7} < 0.05$ ∴ Reject H_0 at $\alpha = 0.05$

There is enough evidence to say that not both stormy and mixed are equal to zero. #

(b)

```
Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
      tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:
    Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
lprice      -0.93014    0.35340  -2.632  0.00977 **
mon         -0.01190    0.20837  -0.057  0.95456
tue         -0.52583    0.20230  -2.599  0.01068 *
wed         -0.56262    0.20696  -2.719  0.00767 **
thu          0.09987    0.20285   0.492  0.62350
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

TABLE 11.5

2SLS Estimates for Fish Demand

Variable	Coefficient	Std. Error	t-Statistic	Prob.
----------	-------------	------------	-------------	-------

<i>C</i>	8.5059	0.1662	51.1890	0.0000
$\ln(PRICE)$	-1.1194	0.4286	-2.6115	0.0103
<i>MON</i>	-0.0254	0.2148	-0.1183	0.9061
<i>TUE</i>	-0.5308	0.2080	-2.5518	0.0122
<i>WED</i>	-0.5664	0.2128	-2.6620	0.0090
<i>THU</i>	0.1093	0.2088	0.5233	0.6018

可觀察到加入工具變數後，彈性估計值的標準誤有變小，P-value 也有變小。

(c)

```
11.24 (c) Sargan 檢定 (查看過度識別有效性):
> summary(model_b, diagnostics = TRUE)

Call:
ivreg(formula = lquan ~ lprice + mon + tue + wed + thu | mon +
      tue + wed + thu + stormy + mixed, data = fultonfish)

Residuals:
    Min       1Q   Median       3Q      Max
-2.1327 -0.4320  0.0654  0.5133  1.1798

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.54023    0.15661  54.533  < 2e-16 ***
lprice      -0.93014    0.35340  -2.632  0.00977 **
mon         -0.01190    0.20837  -0.057  0.95456
tue         -0.52583    0.20230  -2.599  0.01068 *
wed         -0.56262    0.20696  -2.719  0.00767 **
thu          0.09987    0.20285   0.492  0.62350

Diagnostic tests:
      df1 df2 statistic p-value
Weak instruments  2 104    16.140 7.86e-07 ***
Wu-Hausman       1 104     1.489  0.225
Sargan           1  NA     0.772  0.380
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6853 on 105 degrees of freedom
Multiple R-Squared:  0.185,    Adjusted R-squared:  0.1462
Wald test: 4.927 on 5 and 105 DF, p-value: 0.0004317
```

$\therefore$ Sargan value = 0.380 > 0.05 $\therefore$ We can say that The surplus instrument is valid.

(d)

11.24 (d) 星期變數 (MON-THU) 對價格簡化式的聯合顯著性:

```
> f_test_d <- linearHypothesis(model_a, c("mon=0", "tue=0", "wed=0", "thu=0"))  
> print(f_test_d)
```

Linear hypothesis test:

mon = 0

tue = 0

wed = 0

thu = 0

Model 1: restricted model

Model 2: lprice ~ mon + tue + wed + thu + stormy + mixed

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	108	12.408				
2	104	12.115	4	0.29256	0.6279	0.6437

H_0 : mon, tue, wed, thu are equal to zero vs H_1 : not H_0

$\therefore p\text{-value} = 0.6437 > 0.05 \therefore$ Do not reject H_0 at $\alpha=0.05$

There isn't enough evidence to say these variable are not equal to zero.

We can't use these variable to identify supply equation. #

11.28

(a) Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS_i + \alpha_4 DI_i + e_{di}$
 $\Rightarrow P_i = \gamma_1 + \gamma_2 Q_i + \gamma_3 PS_i + \gamma_4 DI_i + e_{di}^*$

預期方向: $\gamma_2 < 0$ (根據需求法則, 斜率為負), $\gamma_3 > 0$ (替代品價格上升帶動需求增加, 進而帶動價格上升)
 $\gamma_4 > 0$ (可支配收入增加帶動價格上升)

Supply: $Q_i = \beta_1 + \beta_2 P_i + \beta_3 PF_i + e_{si}$

$\Rightarrow P_i = \delta_1 + \delta_2 Q_i + \delta_3 PF_i + e_{si}^*$

預期方向: $\delta_2 < 0$ (根據供給法則, 斜率為正), $\delta_3 > 0$ (生產要素價格上升會使供給成本上升, 故供給價格會上升)

(b)

```
--- 2SLS 需求方程結果 ---
> summary(model_demand_2s1s)

Call:
ivreg(formula = p ~ q + ps + di | ps + di + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-39.661  -6.781   2.410   8.320  20.251

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -11.428     13.592   -0.841  0.40810
q             -2.671      1.175   -2.273  0.03154 *
ps             3.461      1.116    3.103  0.00458 **
di            13.390      2.747   4.875  4.68e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.17 on 26 degrees of freedom
Multiple R-Squared: 0.5567,    Adjusted R-squared: 0.5056
Wald test: 17.37 on 3 and 26 DF, p-value: 2.137e-06
```

```
--- 2SLS 供給方程結果 ---
> summary(model_supply_2s1s)

Call:
ivreg(formula = p ~ q + pf | ps + di + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-9.7983 -2.3440 -0.6281  2.4350 11.1600

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -58.7982     5.8592  -10.04 1.32e-10 ***
q             2.9367      0.2158   13.61 1.32e-13 ***
pf            2.9585      0.1560   18.97 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.399 on 27 degrees of freedom
Multiple R-Squared: 0.9486,    Adjusted R-squared: 0.9448
Wald test: 232.7 on 2 and 27 DF, p-value: < 2.2e-16
```

不管是需求還是供給方程式, 係數方向皆符合預期且顯著不等於0 #

(c)

```
> gamma2 <- coef(model_demand_2s1s)["q"]
> p_mean <- mean(truffles$p)
> q_mean <- mean(truffles$q)
> elasticity <- (1/gamma2)*p_mean/q_mean
> cat("平均值處的需求價格彈性為:", elasticity, "\n")
平均值處的需求價格彈性為: -1.272464
```

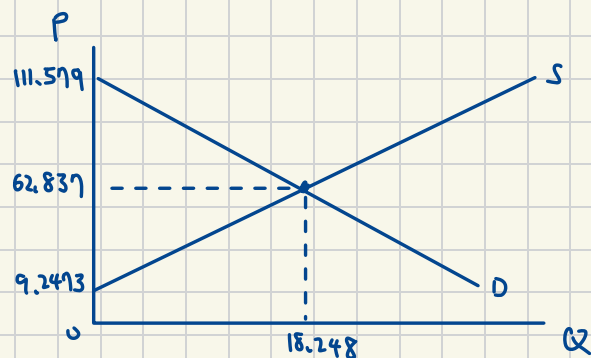
需求價格彈性 $\epsilon_d = \frac{dq}{dp} \cdot \frac{\bar{P}}{\bar{Q}} = \frac{1}{\gamma_2} \cdot \frac{\bar{P}}{\bar{Q}} = -1.272464$ #

(d)

$$\begin{aligned}\text{Demand: } P &= -11.428 - 2.671Q + 3.461 \cdot 22 + 13.39 \cdot 3.5 \\ &= 111.579 - 2.671Q\end{aligned}$$

$$\begin{aligned}\text{Supply: } P &= -58.7982 + 2.9367Q + 2.9585 \cdot 23 \\ &= 9.2473 + 2.9367Q\end{aligned}$$

解聯立得 $Q^* = 18.248$, $P^* = 62.837$



(e)

$$P^* = 62.837, Q^* = 18.248$$

```
> q_star <- 7.8951 + 0.6564*(22) + 2.1672*(3.5) - 0.5070*(23)
> p_star <- -32.5124 + 1.7081*(22) + 7.6025*(3.5) + 1.3539*(23)
> cat("預測均衡價格 P*:", p_star, " 均衡數量 Q*:", q_star, "\n")
預測均衡價格 P*: 62.81425  均衡數量 Q*: 18.2601
```

兩者非常接近,因為 2SLS 本質上就是基於這些簡化式的預測值進行估計。

#

(f)

```
> summary(model_demand_ols)

Call:
lm(formula = p ~ q + ps + di, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-25.0753  -2.7742  -0.4097   4.7079  17.4979

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -13.6195     9.0872  -1.499   0.1460
q             0.1512     0.4988   0.303   0.7642
ps            1.3607     0.5940   2.291   0.0303 *
di            12.3582     1.8254   6.770 3.48e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.814 on 26 degrees of freedom
Multiple R-squared:  0.8013,    Adjusted R-squared:  0.7784
F-statistic: 34.95 on 3 and 26 DF,  p-value: 2.842e-09
```

```
> summary(model_supply_ols)

Call:
lm(formula = p ~ q + pf, data = truffles)

Residuals:
    Min       1Q   Median       3Q      Max
-8.4721  -3.3287   0.1861   2.0785  10.7513

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -52.8763     5.0238  -10.53 4.68e-11 ***
q             2.6613     0.1712   15.54 5.42e-15 ***
pf            2.9217     0.1482   19.71 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.202 on 27 degrees of freedom
Multiple R-squared:  0.9531,    Adjusted R-squared:  0.9496
F-statistic: 274.4 on 2 and 27 DF,  p-value: < 2.2e-16
```

OLS 的估計結果在需求方程上出現了嚴重的正向偏誤，導致係數符號錯誤且不顯著。這證明了在處理具有聯立性質的供需模型時，OLS是失效的，必須採用 2SLS 等工具變數法才能獲得具備經濟一致性的估計值。

#